

HCI4AI course (2024-25)

Orientation



Introduction to the course

Course: Human-Computer Interaction (XB_0013)

Level: First-year Bachelors (AI program)

Credits: 6 ECTs

Instructors:

- Dr. Maryam Alimardani (coordinator)
- Jacopo Michele di Matteo
- Head TA: Lazar Gyoshev

Format: 5 x 1.5h Lectures (Tuesdays/Fridays) + 5 x 1.5h Practical sessions (Monday/Wednesdays, **attendance is mandatory**)

Coursebook: MacKenzie, I. Scott. Human-Computer Interaction: An Empirical Research Perspective

Assessment:

- Written Exam (Individual - 50% of the final grade)
 - Research project report (Group work - 50% of the final grade)
- To pass the course, students must pass both assessment components.**

Eligibility for Assessment:

- "Complete" grade on all 4 assignments that are given throughout the course.
- Resit students cannot transfer any grades from last year.



Course content

Theory:

- What is the history of HCI?
- How can user needs be identified and incorporated into HCI design cycle?
- What are the human factors and cognitive processes involved in HCI?
- How to formulate a valid research question and conduct hypothesis testing?
- What are common research methods to evaluate usability and user experience (UX) in HCI?
- What type of data can be collected in HCI research?
- How to model an interaction?



Practice:

- How to design a human-agent/robot interaction and develop a prototype?
- How to conduct user studies to measure UX and usability of your design?
- How to analyse and report data from HCI research?
- How to write a research paper based on your data?



Course setup

- The course spans over 9 weeks (April–May)
 - Week 1-3 (Lectures)
 - Week 4 (written Exam)
 - Week 6-8 (practical sessions)
 - Week 9 (submission of project report)
- The lectures will give you the theoretical foundation. You are advised to come prepared to the lectures.
- The practical sessions will give you the hands-on skills and research experience. Attendance is mandatory.
- During the practical session, you work in teams on a group project (more on this later)
- The Resit exam is scheduled in June.

Week #	Date & Time	Location	Format	Content	Preparation by students
Week 1	Apr-01 11:00 - 12:45	Theater2- NU-00C745	Lecture	Orientation + HCI History & Human Factors	Read Chapters 1 & 2
	Apr-04 13:30 - 15:15	MF-FG2	Lecture	Interaction Elements & Prototyping	Read Chapter 3
	Apr-04	Assignment 1 is published on April 4 (Submission deadline: Friday, April 11)			
Week 2	Apr-08 11:00 - 12:45	Theater2- NU-00C745	Lecture	Scientific experiments	Read Chapters 4 & 5
	Apr-11 13:30 - 15:15	MF-FG2	Lecture	Hypothesis Testing, Data types & Data analysis	Read Chapter 6
	Apr-11	Assignment 2 is published (Submission deadline: Friday, April 18)			
Week 3	Apr-15 11:00 - 12:45	Theater2- NU-00C745	Lecture	Modeling Interaction, Writing and publishing research papers, Exam Preparation	Read Chapters 7 & 8
Week 4	Apr 23 (15:30-17:45)	Exam			

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- The Resit exam is scheduled in June.

Week 4	Apr 23 (15:30-17:45)	Exam			
Week 6	May-05	Assignment 3 is published (Submission deadline: Monday, May 12)			
	May-07 15:30 - 17:15	differs per team	Practical	Work on Assignment 3	Read everything under <i>Modules</i> > <i>Research projects</i>
Week 7	May 12	Assignment 4 is published (Submission deadline: Monday, May 19)			
	May-12 15:30 - 17:15	differs per team	Practical	Work on Assignment 4	
	May-14 15:30 - 17:15	differs per team	Practical	Work on Assignment 4	
Week 8	May-19 15:30 - 17:15	differs per team	Practical	Work on Project report	
	May-21 15:30 - 17:15	differs per team	Practical	Work on Project report	
Week 9	Deadline May 30 (Friday 23:59)	Submission of Project report			
Week 10	June 03 (18:45 - 21:00)	Resit Exam			

Assignments and submission rules

Week #	Date & Time	Location	Format	Content	Preparation by stu
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Week 4	Apr 23 (15:30-17:45)	Exam			

	May-05	Assignment 3 is published (Submission deadline: Monday, May 12)			
Week 6	May-07 15:30 - 17:15	differs per team	Practical	Work on Assignment 3	Read everything up > Research projects
	May 12	Assignment 4 is published (Submission deadline: Monday, May 19)			
Week 7	May-12 15:30 - 17:15	differs per team	Practical	Work on Assignment 4	
	May-14 15:30 - 17:15	differs per team	Practical	Work on Assignment 4	
Week 8	May-19 15:30 - 17:15	differs per team	Practical	Work on Project report	
	May-21 15:30 - 17:15	differs per team	Practical	Work on Project report	
Week 9	Deadline May 30 (Friday 23:59)	Submission of Project report			
Week 10	June 03 (18:45 - 21:00)	Resit Exam			

Assignments and submission rules

- You will receive 4 assignments in this course:
 - 2 individual assignments (week 1 and 2) including questions related to book chapters
 - 2 group assignments (week 6 and 7) including questions related to your research project.
- **ALL 4 assignments are mandatory and serve as the prerequisite for your exam/project report to be graded.**
- Assignments serve as formative assessment and will only be graded as **complete/incomplete** (no scores).
- You always have 1 week of time to work on each assignment (both individual and group assignments) and submit the report.
- Assignment reports must be prepared using the specific template that will be shared on Canvas. Report must be submitted as **pdf files on Canvas**.
- Submission by email or other communication channels will not be accepted (under no circumstances!)
- **Deadlines are at 23:59 CEST the following week and are observed strictly.**

Assignments and grading rules

- **Late assignment policy:**

If you miss a deadline but still manage to submit a Complete assignment *still within 24 hrs of the deadline*, we will accept it but apply a penalty of 5% (-0.5 points) to your final grade.

After 24hrs, the submission link will disappear and you will no longer be able to submit any documents.



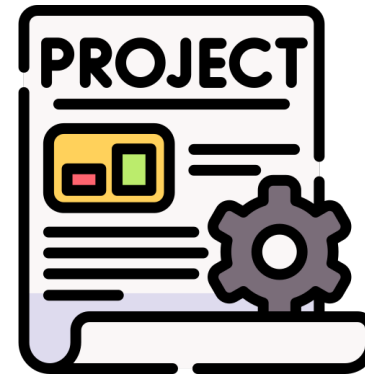
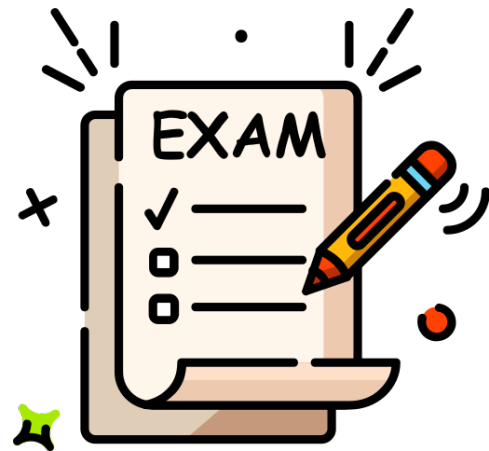
- **Incomplete assignment policy:**

If you submit the assignment on time but your answers to some questions are missing or insufficient, your assignment will be graded "Incomplete". You don't fail the course immediately due to one Incomplete grade. In this case, you will be given a new deadline (max 5 working days) to resubmit a revised version of the report that is complete. However, in this case, we will apply a penalty of 5% (-0.5 points) to your final grade.



Final course grade

Final Grade = 0.5 x Individual Exam + 0.5 x Group Project Report



To pass the course, you should pass both components
(i.e., project report grade ≥ 5.5 , exam grade ≥ 5.5).

Communication with us

If you have a question, **please first check the syllabus**, canvas announcements, and assignment descriptions carefully. It's very likely that your question is already answered there.

- If you have questions about the course or assignment content, and the answer cannot be found on Canvas, then consult your TA during the practical sessions. If the TAs are not able to answer the question, use the Canvas Discussions board to post the question for all instructors and other students/TAs. If your question is still not [adequately] answered, then contact the head TA.

Important note: Please do not send individual emails or canvas messages to instructors. we are not able to read and react to all individual messages.

- To discuss your personal situation and other circumstances that impact your study (e.g., delay in submitting assignments), please contact your academic advisor. Teaching staff are not allowed to have or consider confidential information. The academic advisor will advise the teaching staff.

Practical sessions setup

- Organized in smaller groups of 4 or 5 teams in different classrooms.
- Each group is supervised by a TA who will monitor your progress on the research project.
- The TAs are there to answer your general questions about the assignment questions and group project practicalities; they will not code or troubleshoot for you.
- You must **communicate all of your questions/problems/conflicts/etc. to your TA and not us.**

TA and Room allocation for HCI Teams

TA	Room	Comments	# teams per TA	HCI teams
Mahbod Tajdini	HG-13A36 (23Comp)		5	
Nikolay Pehlivanov	HG-13A37 (24Comp)		5	
Kei Sipani	MF-G406 (30Comp)	Rooms connected	4	
Kevin Caballero-Sellei			4	
Elif Busra Sonmez			5	
Nane Khachaturyan	NU-6A04 (24Comp)		5	
Ella Rozenberg	NU-6B20 (26Comp/24DT)		5	
Atilla Ersezen	NU-6B15 (38Comp/38DT)		5	
Tuana Durmayüksel			5	
Nikolai Bachkaikine	NU-6C33 (24Comp/24DT)		5	
Sofija Kali Kirn	NU-6C40 (22Comp)		5	
Iani Meghea	NU-6C39 (36Comp)		4	
Jose Maria Zuzarte Reis Claver			4	
Lazar Gyoshev			2	

This table is tentative. See the most updated version on Canvas.

Group project context



Human-Robot Interaction

Group project materials

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Practical information	✓	+	⋮
See Syllabus for course manual	✓		⋮
TAs and Room allocation for practical sessions	✓		⋮
Research Project	✓	+	⋮
Designing Human-Robot Interaction with Furhat	✓		⋮
Themes for Group Projects	✓		⋮
Example documents	✓		⋮
Brainstorming with Furhat Consent.docx	✓		⋮
Week 1	✓	+	⋮
Lecture 1	✓		⋮

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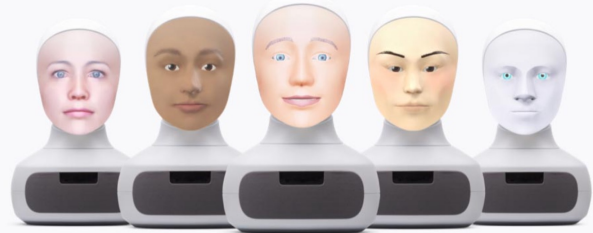
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Designing Human-Robot Interaction with Furhat

The Furhat robot is a social robot developed by the Swedish company Furhat Robotics. It is designed to interact with humans in a natural and conversational way, using a lifelike animated face that can express emotions and engage in dialogue. The Furhat robot is used in various applications such as customer service, education, and research on human-robot interaction.

Due to its design properties, the main components of interaction with the Furhat robot are **speech** and **facial expressions**. Also using the available masks, you can customize different faces for your robot with different ages, genders, and ethnicities (you can even give the robot an [anime character](#) 🤖).



There are various animal- or human-like social robots on the market, some with a full-body design and mobility functions (check e.g., NAO, Pepper, Paro, MiRo, etc.). However, **we chose Furhat for your group project because of the following reasons:**

- 1) We don't have 80 robots for all HCI teams. Furhat Robotics provides the most advanced simulation

Group project themes

1. Education and Learning:

Use social robots in a learning task (e.g., learning a new language). Students can design an interaction with two conditions manipulating robot presence or behavior (e.g., with or without robot, with or without robot gestures, etc.)

2. Healthcare and Therapy:

Use social robots in healthcare (e.g., to provide companionship or emotional support, remind taking medication, perform physical exercise or relaxation techniques, etc.).

3. Customer Service and Hospitality:

Use social robots in customer service roles (e.g., greeting customers at retail stores or providing information at airports, etc.).

Education and Learning

Conversational agents and social robots are often employed in educational contexts to provide personalized and engaging learning experiences for students. These technologies can deliver individualized content and support, answer questions, and provide immediate feedback on students' performance. Additionally, they can stimulate social interactions and collaboration among students, and increase their motivation for learning.

In this project theme, you choose a learning task (for instance learning a new language or a new skill) and design an interaction where the agent/robot is expected to facilitate this learning task for the user. Then you conduct a user study to evaluate the benefit of your interaction design on UX and learning gain. For instance, you can design an interaction with two conditions where you manipulate the robot's facial expressions during a tutoring task (a serious robot vs. a smiley robot) and measure its impact on the learner's engagement and test scores.

Interesting variables to measure in this theme are student engagement with the agent or the task (e.g. [User Engagement Scale](#) ⇨), learning gain and information recall, perception of the robot ([GodSpeed questionnaire](#) ⇨), Acceptance and Use of Technology ([UTAUT questionnaire](#); see [Table 16](#) ⇨), reaction time during the learning or test phases, etc.

Recommended literature:

Belpaeme, T., Kennedy, J., Ramachandran, A., Scassellati, B., & Tanaka, F. (2018). Social robots for education: A review. *Science robotics*, 3(21), eaat5954. ([pdf](#) ⇓)

Khosrawi-Rad, B., Rinn, H., Schlimbach, R., Gebbing, P., Yang, X., Lattemann, C., ... & Robra-Bissantz, S. (2022). Conversational agents in education – a systematic literature review. ([pdf](#) ⇓)

Li, J., Kizilcec, R., Bailenson, J., & Ju, W. (2016). Social robots and virtual agents as lecturers for video instruction. *Computers in Human Behavior*, 55, 1222-1230. ([pdf](#) ⇓)

Saerbeck, M., Schut, T., Bartneck, C., & Janse, M. D. (2010, April). Expressive robots in education: varying the degree of social supportive behavior of a robotic tutor. In *Proceedings of the SIGCHI conference on human factors in computing systems* (pp. 1613-1622). ([pdf](#) ⇓)

Leite, I., Castellano, G., Pereira, A., Martinho, C., & Paiva, A. (2012, March). Modelling empathic behaviour in a robotic game companion for children: an ethnographic study in real-world settings. In *Proceedings of the seventh annual ACM/IEEE international conference on Human-Robot Interaction* (pp. 367-374). ([pdf](#) ⇓)

Shiban, Y., Schelhorn, I., Jobst, V., Hörnlein, A., Puppe, F., Pauli, P., & Mühlberger, A. (2015). The appearance effect: Influences of virtual agent features on performance and motivation. *Computers in Human Behavior*, 49, 5-11. ([pdf](#) ⇓)

Kanero, J., Oranç, C., Koşulu, S., Kumkale, G. T., Göksun, T., & Küntay, A. C. (2022). Are tutor robots for everyone? The influence of attitudes, anxiety, and personality on robot-led language learning. *International Journal of Social Robotics*, 14(2), 297-312. ([pdf](#) ⇓)

Winkle, K., Melsión, G. I., McMillan, D., & Leite, I. (2021, March). Boosting robot credibility and challenging gender norms in responding to abusive behaviour: A case for feminist robots. In *Companion of the 2021 ACM/IEEE international conference on human-robot interaction* (pp. 29-37). ([pdf](#) ⇓)

Calvo-Barajas, N., Perugia, G., & Castellano, G. (2020, August). The effects of robot's facial expressions on children's first impressions of trustworthiness. In *2020 29th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)* (pp. 165-171). IEEE. ([pdf](#) ⇓)

Final project report

- Assignments 3 and 4 will guide you through your project formulation and experimental setup.
- In W9, you submit your final project report + all project materials. **This submission is graded out of 100 and contributes 50% to your final course grade.**
- All group members will receive the same grade.
- A statement of contribution must be included in both assignment (3 and 4) and final reports. **Free-riding will not be tolerated.**

Group Details	
Team number on Canvas:	
Student names and IDs:	
TA name:	
Assignment number:	W#

Question 1.
Answer:

Question 2.
Answer:

...

Statement of Contribution
Examples: A.B. and B.C. carried out the literature review. A.B. prepared the answers to questions X and C.D. D.E. and E.F. prepared the answers to questions Y. F.G. and G.H. programmed the system. H.J. and J.K. created the study protocol. All members helped with the data collection. K.L. and L.M. analyzed the data. M.N. and N.O. created tables and figures. All team members contributed to the writing of the final report.

Study load (take it seriously)

Task	Hours
Attending lectures	2 (x 5 lectures)
Attending practicals	2 (x 5 practicals)
Preparation for the lectures	5 (x 5 lectures)
Working on Assignments	5 (x 4 Assignments)
Research project & report	20 (x 4 weeks)
Preparation for the Final Exam	23
Total	168



Course rules

- Attendance to the practical sessions is mandatory.
 - The TAs will monitor your attendance.
 - You can miss 1 out of 5 sessions without penalty (but please inform your group members and your TA in advance).
 - You should **save this 1 session opportunity for critical situations** such sickness, accidents, train delays, etc.
 - If you miss more than one session without justification, you will lose eligibility for assessment (= fail the course).
 - If due to personal circumstances or health issues you expect to miss more than one practical session, then please consult your situation with the academic advisor ASAP and have them email us.
- Please do not send us questions about the course content, group projects, attendance, etc.
 - Your questions about the assignments, research project and group dynamics should be sent to your TA.
 - Your questions related to personal situations that lead to delays, absence, etc. should be sent to the academic advisor.
 - There is 272 of you and only 2 of us. **We simply cannot and therefore will not answer individual emails!**



Questions?

